Problem 18.15

The present value of an annual perpetuity-immediate of 150 is equal to the present value of an annual perpetuity-immediate that pays 100 at the end of the first 20 years and 200 at the end of year 21 and each year thereafter. Calculate i.

Solution

The present value of the perpetuity-immediate of 150 is

$$150a_{\overline{\infty}|} = \frac{150}{i}.$$

The second perpetuity pays 100 for the first 20 years and 200 thereafter. Its present value is

$$100a_{\overline{20}|} + 200a_{\overline{\infty}|}v^{20}.$$

Equating present values,

$$150a_{\overline{\infty}|} = 100a_{\overline{20}|} + 200a_{\overline{\infty}|}v^{20}.$$

Using

$$a_{\overline{20}|} = \frac{1-v^{20}}{i} \quad \text{and} \quad a_{\overline{\infty}|} = \frac{1}{i},$$

we get

$$\frac{150}{i} = 100\left(\frac{1-v^{20}}{i}\right) + 200\left(\frac{1}{i}\right)v^{20}.$$

Multiplying by i,

$$150 = 100(1-v^{20}) + 200v^{20}.$$

Simplifying,

$$150 = 100 + 100v^{20}.$$

Thus,

$$v^{20} = \frac{1}{2}.$$

Since $v = (1+i)^{-1}$,

$$(1+i)^{-20} = \frac{1}{2}.$$

Therefore,

$$(1+i)^{20} = 2.$$

So

$$1+i = 2^{1/20}.$$

Hence,

$$i = 2^{1/20} - 1 \approx 0.0352649.$$

Therefore,

$$\boxed{i \approx 3.5265\%}.$$